



Sagnik Ghatak

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Robotics, Autonomy & AI Research Engineer

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RESEARCH INTEREST

Structured Grounding of Foundation Models for Long-Horizon Robot Manipulation Planning. Formal Ontological Representations for Dynamic Robot Capability Reasoning and Task Feasibility Verification. Software and systems that make learning-based robot manipulation work in the real world: real-time control of robotic manipulators, machine-learning components for perception and control. Sample-Efficient Reinforcement Learning exploiting Structural Priors for Multi-Agent Autonomous Coordination.

EDUCATION

Technische Hochschule Ingolstadt, Ingolstadt, Germany

Master of Engineering, AI Engineering of Autonomous Systems

Thesis: A Simulation Architecture for Privacy-Preserving Multi-Robot Learning

GPA: 2.1

St. Thomas' College of Engineering and Technology, Kolkata, India

Bachelor of Technology, Electrical Engineering

Thesis: Speed Control of DC Motor Using Fuzzy Logic

CGPA: 8.58/10

PROFESSIONAL EXPERIENCE

AIemotion Bavaria

Wissenschaftlicher Mitarbeiter

Ingolstadt, Germany

04/2026 - 09/2026

- Architecting an end-to-end, web-service-based (REST) **robotic control platform** to support autonomous task execution from natural-language commands, designed for interoperability across tasks and robot platforms.
- Designed and implemented a **two-layer semantic knowledge base** separating the ontological schema (OWL TBox in an RDF triple store, queried via SPARQL) from live world state (ABox in a Redis store), enabling LLM grounding and reasoning without sacrificing real-time write performance.
- Designed and implemented a context-aware **semantic grounding framework** that fuses natural-language input with ontological knowledge and real-time world state to produce structured, machine-interpretable task specifications, grounding LLM intent interpretation in a formal robot knowledge model.
- Designing a **symbolic task planner** that compiles structured intents into verifiable action sequences via PDDL and a domain-independent planner, with ontology-driven capability matching for automatic robot selection and ABox-grounded pose resolution.

Cognizant Technology Solutions

Data Engineer

Kolkata, India

10/2021 - 09/2023

- Large-scale data pipeline development in Python in insurance industry for USA and Canada.

STUDENT EXPERIENCE

AIemotion Bavaria

Studentische Hilfskraft - Robotics

Ingolstadt, Germany

03/2025 - 03/2026

- Deployed a full autonomous navigation stack on TurtleBot4 using **ROS2 and Nav2**, configuring SLAM, costmaps, and recovery behaviours for real-time obstacle avoidance and goal-directed navigation.
- Integrated **MoveIt2** on a Braccio robotic arm to plan and execute collision-free manipulation trajectories, bridging simulation-validated motion plans to physical hardware.
- Developed **Python scripts** to publish goal positions directly to Nav2 and MoveIt2 action servers, enabling repeatable automated task execution without manual RViz input.
- Engineered a **pick-and-place pipeline** on a **UR10e** cobot using **Polyscope** and **OnRobot Eye** for object detection, pose estimation, and closed-loop grasp execution.

PROJECTS

RoboCup Rescue - Unitree Go2 Arm Integration

Robotics Team, THI · MoveIt2, Foxglove, ROS2, Manipulation

Ongoing

- Implemented a **DDS-to-ROS2 middleware bridge** for the Unitree D1 arm using **C++**, exposing joint command and sensor feedback interfaces over **UDP/multicast IPC** with deterministic latency constraints for real-time control within the rescue autonomy stack
- Built a **simulation and visualisation environment** in **Foxglove Studio** for live telemetry monitoring, joint state inspection, and motion debugging during field deployment.

- Implementing a **MoveIt2 motion planning pipeline** for the D1 arm, enabling collision-aware trajectory planning for autonomous manipulation in RoboCup Rescue scenarios.

A Simulation Architecture for Privacy-Preserving Multi-Robot Learning
Master Thesis · IsaacSim, Reinforcement Learning, Federated Learning

2025

- Architected a **Federated Deep Reinforcement Learning (FDRL)** framework integrating IsaacSim, SKRL, and Flower for privacy-preserving multi-agent navigation training across distributed robot teams.
- Implemented and validated **MAPPO** baselines achieving **100% system success rate** and **0% collision rate** in coordinated TurtleBot3 navigation, establishing performance upper bounds for federated comparison.
- Demonstrated that federated MAPPO matches centralised coordination while maintaining data sovereignty, validating decentralised learning viability for real-world multi-robot deployments.

Formula Student Autonomous - State Estimation & Path Planning
Schanzer Racing Electric e.V · ROS2, Python, EKF

03/2024 - 08/2024

- Implemented a **dual Extended Kalman Filter (EKF)** fusing IMU and wheel odometry for high-precision vehicle state estimation under dynamic racing conditions **within an automotive-grade ROS2 pipeline on embedded Linux**.
- Programmed **path planning algorithms** for real-time trajectory generation around track landmarks in a Formula Student Driverless context.

Local Path Planner for Evasive Manoeuvres of Automated Vehicles
Research Project · Python, Vehicle Dynamics, Control Systems

2024

- Developed a local path planner for robust evasive manoeuvres using a **bicycle model** and **PID controller**, improving obstacle avoidance in dynamic environments.

Echo Bot - Autonomous SLAM Robot
ROS2, Python, Gazebo

2024

- Developed an autonomous robot with full **SLAM capabilities** using ROS2, simulated in Gazebo for mapping and navigation validation.

SKILLS

Knowledge Engineering: OWL Ontology Design, Protege, Apache Jena Fuseki, SPARQL

AI & Machine Learning: Deep Learning, Reinforcement Learning, LLM-based Task Planning, Deep Learning

Robotics & Autonomy: Nav2, MoveIt2, SLAM, Navigation, Motion Planning, Manipulation

Simulation: Gazebo, IsaacSim, Foxclove

MLOps & DevOps: Docker, GitHub, CI/CD

Programming: Python, C++, Java, UNIX Scripting

Systems & Middleware: DDS (CycloneDDS/FastDDS), Linux, IPC (pub/sub, shared memory), Ethernet, CAN, ROS2

LANGUAGES

English: C1

German: A2